The Impact of Polysemanticity on the Locality of Neuron-level vs. Layer-level Model Editing

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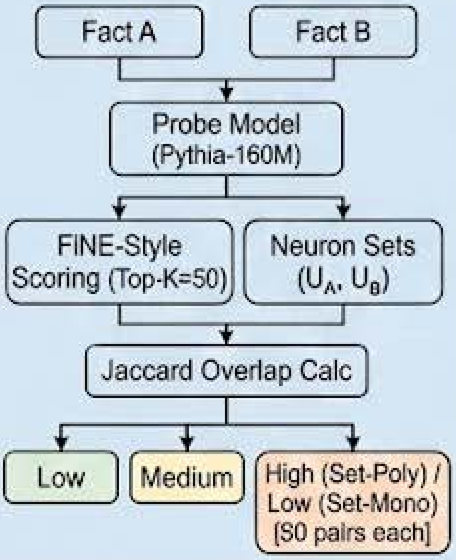
INTRODUCTION & METHODOLOGY

MOTIVATION

Do polysemantic neurons hinder locality? Comparing FiNE (neuron-level) vs. ROME (layer-level).

APPROACH & DATASET



Fact A

Fact B

Probe Model (Pythia-160M)

FiNE-Style Scoring (Top-K=50)

Neuron Sets (U_A, U_B)

Jaccard Overlap Calc

Low

Medium

High (Set-Poly) / Low (Set-Mono) [S0 pairs each]

METRICS

Edit Success (Fact A), Locality (Fact B Acc), Avg. Clipped Drop (Fact B Prob)

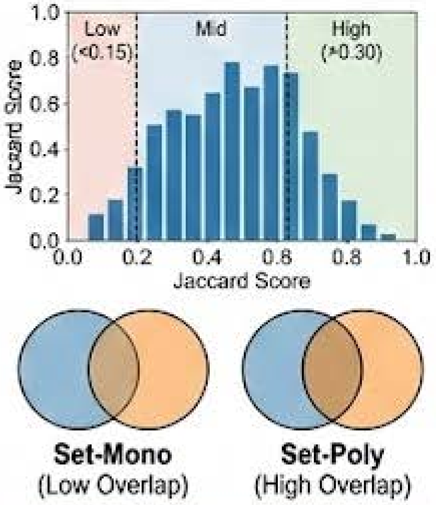
EDITING METHODS (Pythia-6.9B)

Layer-level (ROME): Rank-one update

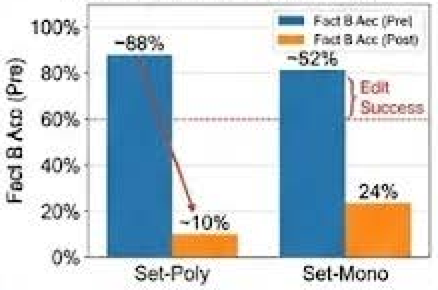
Neuron-level (FiNE): Neuron patch optimization

MAIN RESULTS & ANALYSIS

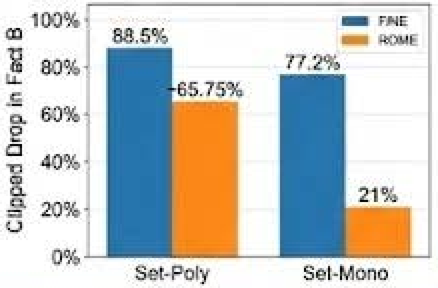
MECHANISTIC OVERLAP REGIMES & DISTRIBUTION



FiNE PERFORMANCE (Set-Poly vs. Set-Mono)



METHOD COMPARISON: AVG. CLIPPED DROP IN FACT B



KEY INSIGHT: High Overlap predicts poor locality for BOTH methods. FiNE is intrinsically "sharper", causing damage even in low-overlap.

DISCUSSION & IMPLICATIONS

KEY FINDINGS

- Polysemantic overlap creates fundamental locality constraints.
- Neuron-level (FINE) is NOT inherently more local.
- FINE is highly sensitive; ROME is more forgiving in low-overlap.

IMPLICATIONS

- Targeting neurons doesn't bypass polysemanticity.
- Need robust, regularized neuron-level objectives.

FUTURE DIRECTIONS

- Scale to larger benchmarks & diverse domains.
- Develop objectives balancing edit strength/locality.

REFERENCES & RESOURCES

ROME: Chen, et al., 2018, IMssometomma'b4yladll), ecanemiodi disnetc; 2023; pythl_a-4-552.
MEND: Chen, et al., 2021. University of Califlemia, 2013. Aumatian et al., 2017. pythi. 4-8678.
FINE: Chen, et al., 2013. Thansucorat of University, Berkeley Vi. 2028: 3050-3866.
Superposition: Chen, st al., 2019. Scanate for owsnemartic monscar sinple analytics, ineryveritlion California, lelitnran; pythi-a-289.

Scan for Code & Data (GitHub)